

Freshdesk API Study Guide

Fetching Tickets, Email Threads, Attachments, Replies & Status Updates

Prepared for ULINK • August 2026

1. Overview

Freshdesk's REST API (v2) exposes tickets, the email/conversation thread on each ticket, attachments, and ticket properties (status, priority, etc.) as JSON resources. Everything ULINK needs — reading a ticket's email, pulling attachments, sending a reply, and updating status — is done through this single API using HTTP Basic Auth.

Base URL for all requests:

```
https://<your-domain>.freshdesk.com/api/v2/
```

Replace <your-domain> with ULINK's Freshdesk subdomain (e.g. ulink.freshdesk.com).

2. Authentication

Freshdesk uses HTTP Basic Auth with your API key as the username and any string (conventionally the letter X) as the password. No OAuth flow is required for account-level API access.

Getting your API key

- Log into Freshdesk → click your profile icon (top right) → Profile Settings.
- Your API key is shown on the right-hand side of that page.
- Treat it like a password — it grants full access to whatever the associated agent account can see.

Example request (curl)

```
curl -u YOUR_API_KEY:X \
-H "Content-Type: application/json" \
https://ulink.freshdesk.com/api/v2/tickets/123
```

Example request (Python)

```
import requests

API_KEY = "YOUR_API_KEY"
DOMAIN = "ulink.freshdesk.com"
auth = (API_KEY, "X")

resp = requests.get(f"https://{DOMAIN}/api/v2/tickets/123", auth=auth)
resp.raise_for_status()
ticket = resp.json()
```

3. Core Concepts

Object	What it represents
Ticket	The support case itself — subject, requester, status, priority, custom fields.

Object	What it represents
Conversation	Each individual email, reply, or forward on a ticket's thread. Fetched separately from the ticket.
Note	An internal, non-customer-facing comment on a ticket (not sent as email).
Attachment	A file object linked to a ticket or a specific conversation entry, with a temporary download URL.

4. Fetching a Ticket and Its Email Thread

To read "the email" on a ticket, you need two things: the ticket object itself (for status/properties) and its conversations (the actual email thread, including the original inbound email and all replies).

4.1 Get a single ticket

```
GET /api/v2/tickets/{id}
```

Add `?include=conversations` to embed the full thread in the same response, e.g.:

```
GET /api/v2/tickets/123?include=conversations
```

4.2 Get the full conversation thread separately

```
GET /api/v2/tickets/{id}/conversations
```

This returns an array of conversation entries in chronological order. Each entry includes `body` (HTML), `body_text` (plain text), `incoming` (true if it came from the customer), `from_email`, `to_emails`, and an `attachments` array.

Python example — fetch ticket + email thread

```
ticket = requests.get(
    f"https://{DOMAIN}/api/v2/tickets/123",
    auth=auth
).json()

conversations = requests.get(
    f"https://{DOMAIN}/api/v2/tickets/123/conversations",
    auth=auth
).json()

for msg in conversations:
    print(msg["incoming"], msg["from_email"], msg["body_text"][:80])
```

Note: conversations, like most list endpoints, are paginated — see Section 8.

5. Fetching Attachments

Attachments live inside the ticket object and inside each conversation entry, never as a standalone "list attachments" endpoint. Each attachment object looks like this:

```
{
  "id": 55000012345,
  "name": "invoice.pdf",
  "content_type": "application/pdf",
  "size": 102450,
  "attachment_url": "https://freshdesk-attachments.s3.amazonaws.com/...",
  "created_at": "2026-08-01T09:12:00Z"
}
```

Where to find them

- Ticket-level attachments (files attached directly, not via email): `ticket.attachments`
- Per-email attachments: each item in the `conversations` array has its own `.attachments` list

Downloading a file

`attachment_url` is a pre-signed S3 URL that is valid for a limited time (roughly 24 hours). No Freshdesk auth header is needed to download from it — just GET the URL directly and stream the bytes to disk.

```
att = conversations[0]["attachments"][0]
file_resp = requests.get(att["attachment_url"])
with open(att["name"], "wb") as f:
    f.write(file_resp.content)
```

Because the URL expires, always download attachments soon after fetching the ticket/conversation — don't store the URL long-term as a reference.

6. Replying to a Ticket (Sending an Email)

A reply is sent to the requester as an actual outbound email and is appended to the ticket's conversation thread.

POST `/api/v2/tickets/{id}/reply`

Request body

Field	Description
<code>body</code>	Required. HTML content of the reply.
<code>cc_emails</code>	Optional array of CC addresses.
<code>bcc_emails</code>	Optional array of BCC addresses.
<code>from_email</code>	Optional — use if the ticket's product supports multiple support emails.
<code>attachments</code>	Optional, multipart/form-data only (see note below).

Python example

```
payload = {
  "body": "Hi, thanks for reaching out — we've resolved this on our end."
}

resp = requests.post(
  f"https://{DOMAIN}/api/v2/tickets/123/reply",
  auth=auth,
```

```
    json=payload
)
resp.raise_for_status()
```

If you need to attach a file to your reply, send the request as multipart/form-data instead of JSON (use requests' files= parameter), since the API can't accept binary content inside a JSON body.

For an internal-only comment instead of a customer-facing reply, use POST /api/v2/tickets/{id}/notes with the same body field.

7. Updating Ticket Status (and Other Properties)

PUT /api/v2/tickets/{id}

Send only the fields you want to change — Freshdesk merges them into the existing ticket.

Python example — mark a ticket Resolved

```
payload = {"status": 4} # 4 = Resolved

resp = requests.put(
    f'https://{DOMAIN}/api/v2/tickets/123',
    auth=auth,
    json=payload
)
resp.raise_for_status()
```

The same endpoint updates priority, group_id, responder_id (assigned agent), type, and custom_fields, e.g.:

```
{
  "status": 3,
  "priority": 2,
  "custom_fields": {
    "cf_order_number": "UL-88213"
  }
}
```

Default status values

ID	Status
2	Open
3	Pending
4	Resolved
5	Closed

Accounts can add custom statuses beyond 5; if ULINK has custom statuses configured, fetch GET /api/v2/ticket_fields to see the actual numeric values in use.

Default priority values

ID	Priority
1	Low

ID	Priority
2	Medium
3	High
4	Urgent

8. Rate Limits, Pagination & Errors

Rate limits

- Limits are enforced per account per hour, based on your Freshdesk plan.
- Every response includes X-Ratelimit-Total and X-Ratelimit-Remaining headers — check these to avoid surprises.
- Exceeding the limit returns HTTP 429; back off and retry after the window resets.

Pagination

- List endpoints (e.g. GET /tickets) accept page and per_page (max 100) query params.
- Unfiltered ticket listing only returns up to 300 pages of results — for full history syncs, filter by updated_since instead of paging through everything.

Common error codes

Code	Meaning
400	Bad request — malformed JSON or invalid field value.
401	Invalid API key/credentials.
403	Valid credentials, but no permission for this action/ticket.
404	Ticket or resource not found.
429	Rate limit exceeded.

9. End-to-End Example Workflow

A minimal script that fetches a ticket's email thread and attachments, replies, then closes it:

```
import requests

DOMAIN = "ulink.freshdesk.com"
auth = ("YOUR_API_KEY", "X")
ticket_id = 123
base = f"https://{DOMAIN}/api/v2/tickets/{ticket_id}"

# 1. Fetch ticket + email thread
ticket = requests.get(base, auth=auth).json()
convo = requests.get(f"{base}/conversations", auth=auth).json()

# 2. Download any attachments
for msg in convo:
    for att in msg.get("attachments", []):
        data = requests.get(att["attachment_url"]).content
        open(att["name"], "wb").write(data)

# 3. Send a reply
```

```
requests.post(f"{base}/reply", auth=auth,  
             json={"body": "Thanks — we've reviewed the attached files and resolved this."}  
             ).raise_for_status()
```

```
# 4. Update status to Resolved  
requests.put(base, auth=auth, json={"status": 4}).raise_for_status()
```

10. Reference Links

[Freshdesk API documentation](#)

[Freshworks Developer Docs — REST APIs](#)

[Practical guide to the Freshdesk ticket API](#)

[Practical guide to the Freshdesk Conversations API](#)